F1

Como $\cos\alpha = \dfrac{a_x}{a}$ y $\cos\beta = \dfrac{a_y}{a}$

$\cos 60° = \dfrac{a_x}{a}$; $\cos 60° = \dfrac{a_y}{a}$

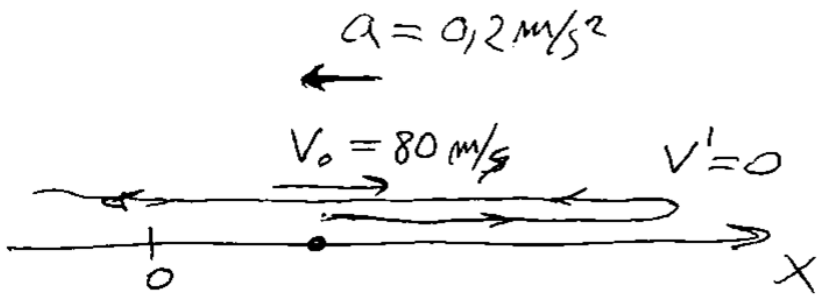
$$\Rightarrow a_x = a_y = a\cos 60°$$
$$= 10\underbrace{\cos 60°}$$
$$= 10\left(\frac{1}{2}\right)$$

$$a_x = a_y = \underline{5} //$$

$$\therefore \quad a^2 = a_x^2 + a_y^2 + a_z^2$$
$$a_z = \sqrt{a^2 - a_x^2 - a_y^2}$$
$$a_z = \sqrt{10^2 - 5^2 - 5^2}$$
$$a_z = 5\sqrt{2}$$

$$\Rightarrow \underline{\vec{a} = (5, 5, 5\sqrt{2})} //$$

F2



$$V' = V_0 + at$$

$$0 = 80 - 0{,}2\,t'$$

$$t' = t_{ida} = \frac{-80}{-0{,}2} = 400\,(s)$$

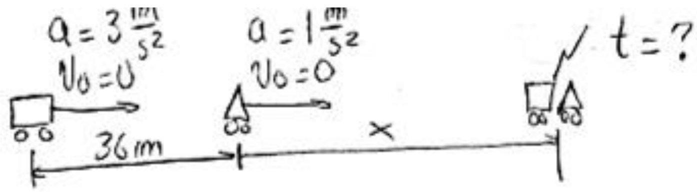
$$t_{ida} = t_{vuelta}$$

Como el $t_{total} = t_{ida} + t_{vuelta}$

$$= 400 + 400$$

$$t_{total} = 800\,(s)$$

F3



Para el auto

$\rightarrow d = V_0 \cdot t + \frac{1}{2} \cdot a \cdot t^2$

$x + 36 = 0 \cdot t + \frac{1}{2} \cdot 3 \cdot t^2$

$x = \frac{3}{2} t^2 - 36 \ldots Ec1$

Para la moto

$\rightarrow d = V_0 \cdot t + \frac{1}{2} \cdot a \cdot t^2$

$x = 0 \cdot t + \frac{1}{2} \cdot 1 \cdot t^2$

$x = \frac{t^2}{2} \ldots Ec2$

$Ec1 = Ec2$

$\frac{3}{2} t^2 - 36 = \frac{t^2}{2} \quad (\times 2)$

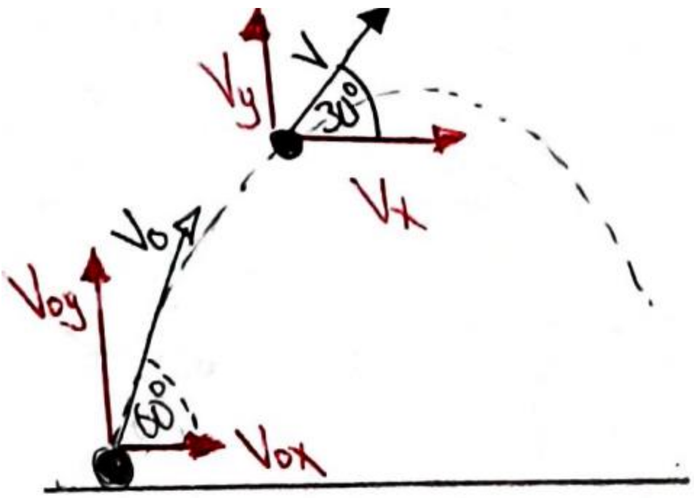
$3t^2 - 72 = t^2$

$2t^2 = 72$

$\sqrt{t^2} = \sqrt{36}$

$\boxed{t = 6 \, [s]}$

F4



$$V_{ox} = V_o \cos 60 = 10$$

$$V_{oy} = V_o \,\text{sen} 60 = 10\sqrt{3}$$

"x"

$$V_{ox} = V_x$$

$$10 = V \cos 30$$

$$V = \frac{20}{3}\sqrt{3}$$

"y"

$$V_{Fy} = V_{oy} - gt$$

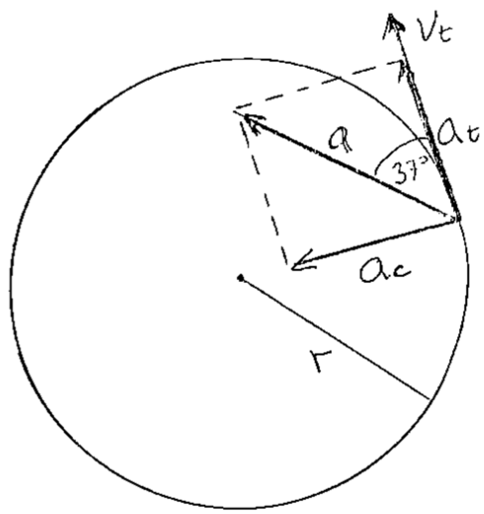
$$V \,\text{sen} 30 = 10\sqrt{3} - 10t$$

$$\frac{20\sqrt{3}}{3} \cdot \frac{1}{2} = 10\sqrt{3} - 10t$$

$$-\frac{20\sqrt{3}}{3} = -10t$$

$$1.2\,[s] = t$$

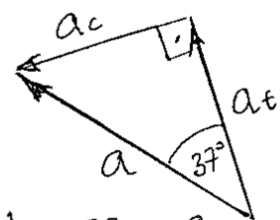
F5



$$a_c = \frac{V_t^2}{r}$$

$$a_t = r\alpha$$

$$a = \sqrt{a_t^2 + a_c^2}$$



$$tg\,37° = \frac{a_c}{a_t}$$

Como: $a_c = \frac{V_t^2}{r}$ y $a_t = r\alpha \leftarrow \quad a_c = tg\,37°\, a_t$

$$\Rightarrow \quad \frac{V_t^2}{r} = tg\,37° (r\alpha)$$

$$V_t^2 = (tg\,37°)\, r^2 \alpha \quad (1)$$

Como $\omega_0 = 0 \Rightarrow \omega = \cancel{\omega_0}^{0} + \alpha t$

$$\underline{\omega} = \alpha t$$

$$\therefore \quad V_t = r\underline{\omega}$$

$$V_t = r\alpha t$$

$$V_t^2 = r^2 \alpha^2 t^2 \quad (2)$$

(1) en (2)

$$(tg\,37°)\cancel{r^2}\cancel{\alpha} = \cancel{r^2}\alpha^{\cancel{2}} t^2 \quad \div\alpha\ (\alpha \neq 0)$$

$$\alpha t^2 = tg\,37°$$

Para $t = 0{,}5(s)$: $\alpha (0{,}5)^2 = tg\,37°$

$$\alpha \cong \frac{0{,}75}{0{,}25}$$

$$\underline{\alpha = 3{,}0\ rad/s^2}$$